

Solution. Let

$$P(x) = x^7 + a_6x^6 + a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0 \quad (a_i \in \mathbb{Z})$$

be a *good* polynomial. For $i = 0, \dots, 6$ put

$$c_i := |a_i| \ (> 0), \quad c_7 := |a_7| = 1.$$

All the numbers c_i are odd, satisfy $c_i < 2025^2$ and, by definition of a good polynomial,

$$P(c_i) = 0 \quad (i = 0, \dots, 7).$$

0.0.1 1. Preliminary observations

Because the leading coefficient is 1, evaluating at $x = 1$ gives

$$P(1) = 1 + a_6 + \dots + a_0 = 0 \implies a_6 + \dots + a_0 = -1.$$

Since each a_i is odd, (2) implies that among the seven numbers $a_0, \dots, a_6$ exactly three are $+1$ and four are -1 . Consequently every c_i equals either 1 or $|a_i| > 1$.

0.0.2 2. The largest absolute coefficient

Let

$$r := \max\{c_0, c_1, \dots, c_6\}.$$

If every $c_i = 1$ we shall deal with this case in §4. Hence assume from now on that $r > 1$.

Lemma 2.1 $c_0 = r$ and c_0 is the absolute value of the product of all roots of P .

Proof. Write the factorisation of P by its roots $\alpha_1, \dots, \alpha_7$:

$$P(x) = \prod_{j=1}^7 (x - \alpha_j), \quad \alpha_j \in \mathbb{C}.$$

The constant term is

$$a_0 = (-1)^7 \prod_{j=1}^7 \alpha_j = - \prod_{j=1}^7 \alpha_j.$$

Hence

$$c_0 = |a_0| = \left| \prod_{j=1}^7 \alpha_j \right|.$$

All numbers c_i are roots of P by (1), thus 1 (the absolute value of the leading coefficient) is a root. Consequently $\prod_{j=1}^7 \alpha_j$ is a product of numbers each of which has absolute value at least 1; therefore the right-hand side of (3) is at least the largest of the numbers $|\alpha_j|$. By (1) every $|\alpha_j|$ belongs to the set $\{c_0, \dots, c_6\}$; hence $\max_j |\alpha_j| = r$. Thus $c_0 \geq r$. On the other hand c_0 itself belongs to the set $\{c_0, \dots, c_6\}$; therefore $c_0 \leq r$. Consequently $c_0 = r$.

0.0.3 3. Reduction to a quadratic factor

Because of $P(1) = 0$ (see (2)), the linear factor $(x - 1)$ divides P . By Lemma 2.1 the integer $r = c_0$ is a root of P ; hence $(x - r)$ also divides P . Write

$$P(x) = (x - 1)(x - r)Q(x),$$

where Q is monic of degree 5 and has integer coefficients. Put

$$Q(x) = x^5 + b_4x^4 + b_3x^3 + b_2x^2 + b_1x + b_0, \quad b_j \in \mathbb{Z}.$$

Expanding (4) and comparing coefficients with the generic form of P yields

$$\begin{aligned} a_6 &= b_4 - (1 + r), \\ a_5 &= b_3 - (1 + r)b_4 + r, \\ a_4 &= b_2 - (1 + r)b_3 + rb_4, \\ a_3 &= b_1 - (1 + r)b_2 + rb_3, \\ a_2 &= b_0 - (1 + r)b_1 + rb_2, \\ a_1 &= -(1 + r)b_0 + rb_1, \\ a_0 &= r b_0. \end{aligned}$$

Recall that $c_i = |a_i|$ equals either 1 or r ; the latter is the maximal value. From the last line of (5),

$$c_0 = |a_0| = |r b_0|.$$

Since $c_0 = r > 1$, the only possibility is

$$|b_0| = 1 \quad \text{and} \quad c_0 = r.$$

Let us write $b_0 = \varepsilon_0$ with $\varepsilon_0 \in \{\pm 1\}$.

Lemma 3.1 $c_1 = 1$ and $b_1 = -\varepsilon_0$.

Proof. From (5) and (6),

$$a_1 = -(1 + r)\varepsilon_0 + rb_1, \quad c_1 = |a_1| \leq r.$$

If $c_1 = r$ then

$$r = |-(1+r)\varepsilon_0 + rb_1| = |r(b_1 - \varepsilon_0) - \varepsilon_0|.$$

Dividing by r gives $|b_1 - \varepsilon_0 - \varepsilon_0/r| = 1$. Because $r > 1$ the fraction ε_0/r is not integral, hence the equality is impossible. Thus $c_1 \neq r$; consequently $c_1 = 1$. Now $|-(1+r)\varepsilon_0 + rb_1| = 1$ implies $rb_1 = (1+r)\varepsilon_0 \pm 1$. Since r does not divide the integer ± 1 (again because $r > 1$), the only way to obtain an integer b_1 is

$$b_1 = -\varepsilon_0.$$

Lemma 3.2 No odd integer $r > 1$ satisfies the remaining equations (5).

Proof. Insert $(b_{-0} = -\varepsilon_0)$ and $(b_1 = -\varepsilon_0)$ from (6)–(7) into the expression for (a_2) in (5):

$$a_2 = \varepsilon_0 - (1+r)(-\varepsilon_0) + rb_2 = rb_2 + \varepsilon_0(r+2).$$

Hence

$$c_2 = |a_2| = |rb_2 + \varepsilon_0(r+2)| = |r(b_2 + \varepsilon_0) + 2\varepsilon_0|.$$

Because the absolute values of the coefficients may only be 1 or r , we must have $c_2 \in \{1, r\}$.

Case $c_2 = 1$. Then $(|r(b_{-2} + \varepsilon_0) + 2\varepsilon_0| = 1)$. The term $(r(b_{-2} + \varepsilon_0) + 2\varepsilon_0)$ is a multiple of $r \geq 3$ and therefore its absolute value is at least 3 unless it is zero. If it is zero we obtain $|2\varepsilon_0| = 2 \neq 1$. Hence this case is impossible.

Case $c_2 = r$. Now $|r(b_2 + \varepsilon_0) + 2\varepsilon_0| = r$, i.e.

$$|b_2 + \varepsilon_0 + \frac{2\varepsilon_0}{r}| = 1.$$

Since $0 < |\frac{2\varepsilon_0}{r}| < 1$ (recall $r \geq 3$), the left-hand side can be equal to 1 only when $b_2 + \varepsilon_0 = 0$ and $\frac{2\varepsilon_0}{r} = 0$, the latter being impossible for $r > 1$. Thus this case also leads to a contradiction.

Both possibilities for c_2 are impossible; therefore no odd $r > 1$ can satisfy the system (5).

From Lemmas 3.1 and 3.2 we deduce that the assumption $r > 1$ is false. Consequently

$$\boxed{c_0 = c_1 = \dots = c_6 = 1}.$$

That is, **every coefficient of a good polynomial has absolute value 1**. Hence each coefficient is either $+1$ or -1 .

0.0.4 4. Determining all good polynomials

Because all $|a_i| = 1$, condition (1) reduces to the single requirement that 1 be a root of P ; indeed $P(1) = 0$ is equivalent to

$$1 + a_6 + \cdots + a_0 = 0.$$

Let p be the number of coefficients equal to $+1$ among $a_0, \dots, a_6$; then the remaining $7 - p$ coefficients equal -1 . Equation (11) becomes

$$1 + (p - (7 - p)) = 0 \iff 2p - 6 = 0 \iff p = 3.$$

Thus exactly three of the seven coefficients $a_0, \dots, a_6$ are $+1$ and the other four are -1 . Choosing the three positions of the $+1$'s determines the polynomial uniquely; there are

$$\binom{7}{3} = 35$$

such choices.

Every polynomial obtained in this way is monic of degree 7, has all coefficients odd and smaller than 2025^2 , and satisfies $P(1) = 0$; because each absolute coefficient equals 1, condition (1) is fulfilled. Hence all 35 polynomials are good.

0.0.5 5. Conclusion

The only possible good polynomials are those whose coefficients are ± 1 with exactly three $+1$'s and four -1 's; there are

$$\boxed{35}$$

such polynomials.